1's Complement of 10011 : 01100

$$\begin{array}{r} 10010 \\ 01100 \\ \hline 11110 \end{array}$$

1's Complement of 11110 : 00001

$\boxed{\text{Ans : } -1}$

d) 100 − 110000

M = 000100
N = 110000

2's complement of N :

$$\begin{array}{r} 001111 \\ + \quad 1 \\ \hline 010000 \end{array}$$

$$\begin{array}{r} 000100 \\ +010000 \\ \hline 010100 \end{array} \rightarrow \text{2's complement : } \begin{array}{r} 101011 \\ + \quad 1 \\ \hline 101100 \end{array}$$

Ans : −101100

1's complement of N : 001111

$$\begin{array}{r} 000100 \\ +001111 \\ \hline 010011 \end{array} \rightarrow \text{1's complement : } \boxed{-101100}$$

1.8 Binary Logic

1.8.1 Definition of Binary Logic

Binary logic consists of binary variables and logical operations. The variables are designated by letters of the alphabet such as A, B, C, x, y, z etc, with each variable having two and only two distinct possible values : 1 and 0. There are three basic logical operations : AND, OR and NOT.

1. $\boxed{\text{AND :}}$ This operation is represented by a dot or by the absence of an operator. For example, $x.y = z$ or $xy = z$ is read "x AND y is equal to z." The logical operation AND is interpreted to mean that $z=1$ if and only if $x=1$ and $y=1$; otherwise $z=0$.
2. $\boxed{\text{OR :}}$ This operation is represented by a plus sign. For example, $x+y=z$ is read "x OR y is equal to z." meaning that $z=1$ if $x=1$ or if $y=1$ or if both $x=1$ and $y=1$. If both $x=0$ and $y=0$, then $z=0$.
3. $\boxed{\text{NOT :}}$ This operation is represented by a prime (sometimes by a bar). For example, $x'=z$ (or $\bar{x}=z$) is read "x not is equal to z," meaning that z is what x is not. In other words, if $x=1$, then $z=0$ but if $x=0$, then $z=1$.